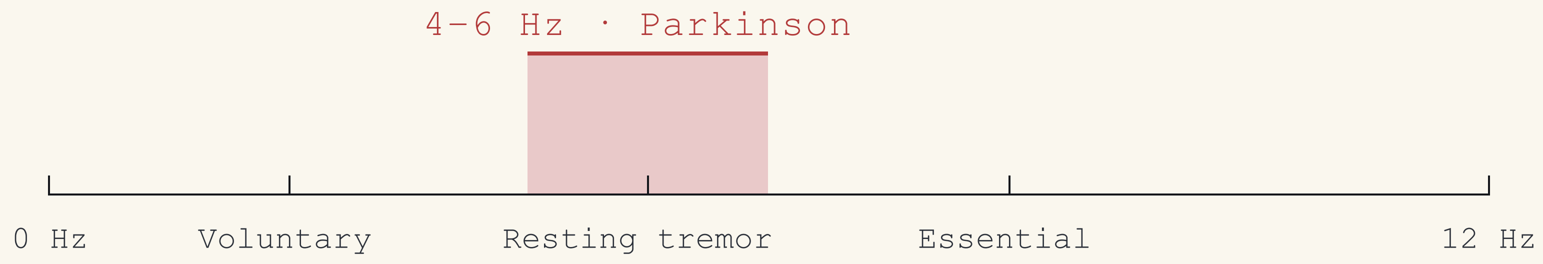
RESEARCH QUESTION

§ 1.0

Can a consumer smartphone's built-in accelerometer reliably detect the characteristic *4–6 Hz resting tremor* associated with Parkinson's disease?

FIG. 1 — TREMOR FREQUENCY TAXONOMY

The diagnostic band is narrow and well separated from voluntary motion.



H_0 — WORKING HYPOTHESIS

The acceleration magnitude spectrum exhibits a dominant peak inside the 4–6 Hz band when, and only when, parkinsonian tremor is present.

INSTRUMENTATION

A single MEMS accelerometer, *three axes*, no add-ons.

WHY AN ACCELEROMETER?

i **Prevalence .** Already present in every modern smartphone, no extra hardware, no compliance burden.

ii **Sensitivity.** MEMS resolution easily resolves the sub-centimetre displacements of resting tremor.

iii **Validated.** Prior literature establishes pocket-mounted accelerometry as viable for tremor monitoring.

DEVICE UNDER TEST

ID DUT-01

REV 1.0

2025

Samsung Galaxy S24 Ultra

SM-S928B · Android 14

SENSOR	STMicroelectronics LSM6DSV 6-axis IMU (accelerometer + gyroscope), non-wake-up mode
SAMPLING RATE	≈ 116 Hz Nyquist 58 Hz — well above the 4–6 Hz band of interest
QUANTITY	Linear acceleration in m/s^2 Three orthogonal body-fixed axes (X, Y, Z); gravity included, removed in pre-processing

X · LATERAL

Side-to-side acceleration of the device chassis.

Y · LONGITUDINAL

Along the phone's long edge — gravity-aligned in portrait.

Z · NORMAL

Perpendicular to the screen — pocket-facing axis.

ANALYTICS & RESULT

From time series to a *frequency verdict.*

01 **Acquire X / Y / Z**
3-axis recording at ~116 Hz, ≈ 420 samples (3.6 s).

02 **Magnitude & detrend**
$\|a\| = \sqrt{x^2 + y^2 + z^2}$; subtract running mean.

03 **FFT spectrum**
DFT of detrended magnitude; inspect 4–6 Hz band.

FINDING · HEALTHY SUBJECT

No tremor detected

No significant peak in 4–6 Hz. Energy concentrates *below 2 Hz* , consistent with brief voluntary movement rather than parkinsonian oscillation.

Fig. 2 — Raw 3-axis acceleration over time

TIME DOMAIN

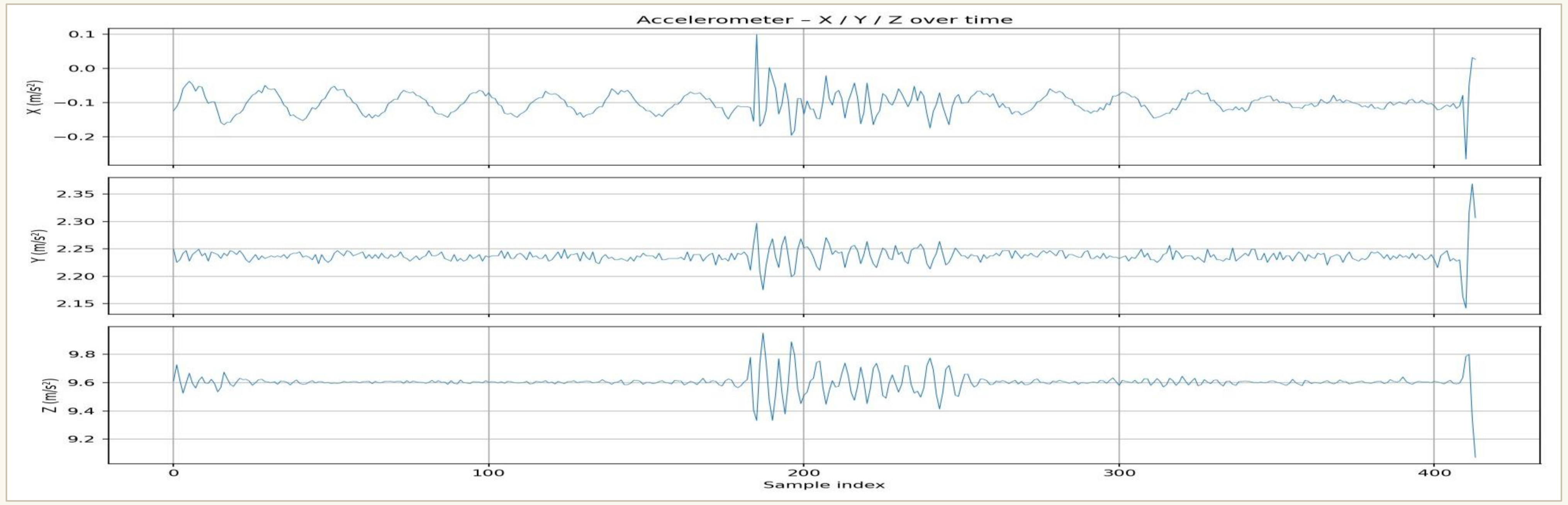
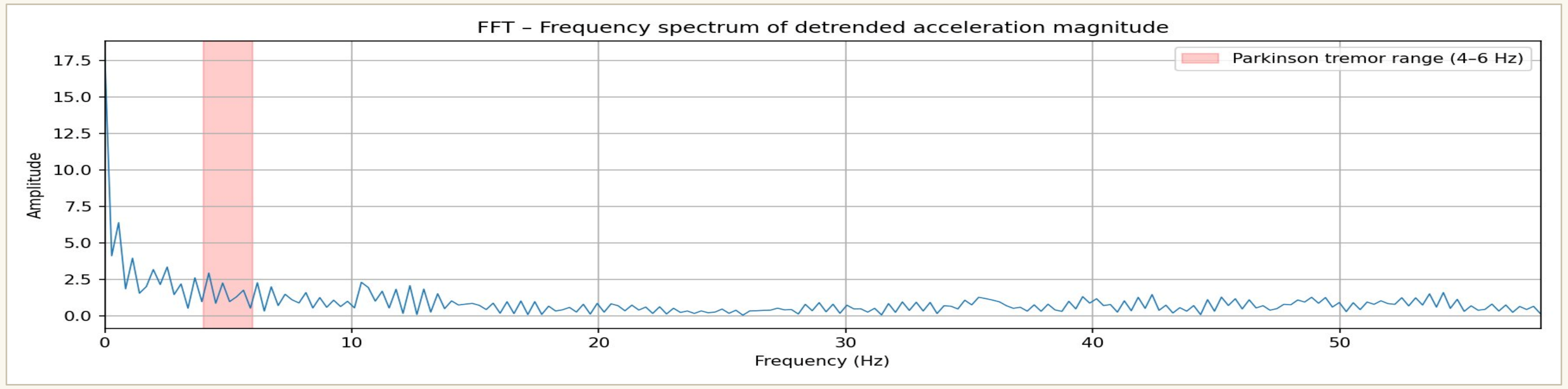


Fig. 3 — FFT spectrum, Parkinson band highlighted

FREQUENCY · RESULT



Fachhochschule Nordwestschweiz